

Customer Sentiment Analysis & Review Mining for E-Commerce Apparel

Business Text Analytics Capstone — MBA, GITAM School of Business (GSB)



Executive Summary

We built an automated NLP pipeline to transform unstructured product reviews into actionable insights for apparel retailers. The solution classifies sentiment, surfaces recurring product defects, and recommends targeted operational fixes to reduce returns and improve catalog satisfaction.

Business Problem & Objectives

Problem

E-commerce apparel sites receive high volumes of free-text reviews; manual analysis is slow and inconsistent, obscuring root causes of returns.

Core Gap

Star ratings quantify dissatisfaction but do not explain whether returns are due to fit, fabric, or expectation mismatch.

Objective

Design a reliable Python NLP pipeline to classify sentiment and detect recurring issues (fit, fabric, sizing, imagery mismatch).

Target Outcome

Lower reverse logistics costs and return rates by prioritizing corrective actions in product pages, QA, and support routing.

Dataset & Preprocessing

Source: Women's E-Commerce Clothing Reviews (Kaggle)— 23,486 entries across 10 fields.

Women's ECommerce Clothing Reviews Despamnicte

Rating	Department	Volume	Rating	Matiny	Babbing
Review Text	00.11M		1/2.33	1	
Review Text	Department Name	\$30060 (7.01)	\$830.558	⊙	
Review Text	1500 340			⊙	
Rating Department	Moist	Momon Boudry \$210...	Afeg+ex dovameandoneil te caone first 4)		
Review Text	Obvlesse Ecommoond	Beatunie thice (Loicolt)	ATNY5396	5	
Rating Department	007.628				
Devevns		⊙			
Review Text	Dustlireng	Natind	ATNY5.96		
Rating Department	Retting	Gfcszebr	\$32932		
Necloy OnradahlMame	Clommene Srahhting	rmoocinen afobll	Propfersemcer (decktchcs)	< Monopheedhal ⊙	
			6/96.38 0.9021		



- Target labeling: Ratings 4–5 → Positive; 1–2 → Negative; 3 removed to avoid ambiguity.
- Text cleaning: lowercase, regex to remove numbers/punctuation, stopwords filtered with NLTK, lemmatized via WordNetLemmatizer.
- Features: Primary text field, rating, department, and metadata used for segment analysis.

Exploratory Text Analysis

Key observations:

- Distribution skew: majority positive reviews – common in post-purchase feedback.
- Positive drivers: comfort, fit, fabric softness, flattering cut, value perception.
- Negative drivers: sizing mismatches (words like "huge", "tight"), perceived cheap materials, and imagery mismatch leading to disappointment.

Visuals: side-by-side word clouds and frequency histograms highlight polarity drivers for product teams.

Modeling Approach



Pipeline highlights:

- Feature extraction: TF-IDF limited to top 5,000 tokens to balance signal and speed.
- Model: Logistic Regression on stratified 80/20 train/test split for reproducibility and interpretability.
- Post-processing: Threshold tuning and confidence scoring to surface high-priority negative cases.

Performance Metrics

Key metrics (binary Positive vs Negative):

93%

Accuracy

0.93

Positive Precision

Recall 0.99 | F1 0.96

0.83

Negative Precision

Recall 0.49 | F1 0.61

—

Unseen validation

Model flags negatives with avg confidence ~95.5% and positives ~99.0% for prioritized review handling.

Business Insights & Recommendations



1. Size & Fit Standardisation

High frequency of “huge” and “unflattering” indicates inconsistent sizing. Implement standardized size charts, 3-D fit tools, and user height/weight guides to reduce fit-related returns.



2. Material Quality Controls

Mentions of “cheap” and “disappointed” point to fabric and imagery mismatch. Strengthen batch QA and update product imagery with fabric close-ups and honest descriptions.



3. Automated Service Routing

Integrate classifier into support portal to auto-flag urgent negatives for retention workflows; route fit issues to product/merch teams and quality complaints to suppliers.

Limitations

Current constraints:

- Class imbalance lowered negative recall (0.49); subtle or sarcastic language remains challenging.
- Binary labels remove neutral nuance — future work should support multi-class or continuous sentiment and aspect tagging.
- Operational integration requires A/B testing to quantify return-rate improvements and ROI.

Conclusion

The NLP model achieved 93% accuracy in sentiment classification, turning customer reviews into clear, actionable insight.

It identified fit, fabric, and imagery mismatch as the main drivers of negative feedback.

Terms like "huge" and "cheap" were strongly linked to dissatisfaction.

The results provide a practical foundation for product, marketing, and customer service improvements.

Overall, the project shows how text analytics can reduce returns, improve customer satisfaction, and support smarter business decisions.

Thank You

Kumudhini – 2025337529

Shreya – 2025027663

Neeraj – 2025008753

Meghana – 20225724130

Likhitha – 2025679727